

Yuan Liao

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Google Scholar: Citations 196, h-index 7, i10-index 6 | GitHub
The Chinese University of Hong Kong, Shenzhen

EDUCATION

Nanjing University of Posts and Telecommunications

M.Eng. in Circuits and Systems; Double First-Class Discipline (A-)

Nanjing University of Posts and Telecommunications

B.Eng. in Automation

Aug. 2022 – Jun. 2025

GPA: 3.93/5.00; Rank: 6/30

Aug. 2018 – Jun. 2022

GPA: 3.50/5.00; Rank: 18/149

RESEARCH EXPERIENCE

The Chinese University of Hong Kong, Shenzhen

Nov. 2024 – Present

Research Assistant

Shenzhen, China

- **Supervisors:** Prof. Siqi Cai, Prof. Haizhou Li (IEEE Fellow)
- **Activity:** Focused on auditory brain-computer interfaces for brain-controlled hearing aids. Key contributions include proposing SincAlignNet for EEG-audio alignment, FD-ARL for cross-subject auditory attention decoding, and BM-TSE for brainprint-modulated target speaker extraction.
- **Outcomes:** One first-author manuscript under revision at *IEEE Journal of Biomedical and Health Informatics* and two ICASSP 2026 papers published in IEEE Xplore.

State Key Laboratory of Flexible Electronics, NJUPT

May 2022 – Present

Graduate Research Assistant

Nanjing, China

- **Supervisors:** Prof. Liya Huang, Prof. Qiang Zhao
- **Activity:** Conducted cross-disciplinary research on flexible neural interfaces, wearable physiological sensors, and EEG-based BCI applications. My role involved designing BCI experimental paradigms, conducting human-subject experiments, and developing deep learning models for neural and physiological signal decoding.
- **Outcomes:** One equal-contribution co-first-author paper published in *Nature Sensors*; three published papers in *Matter*, *ACS Sensors*, and *Science China Information Sciences*; and one first-author manuscript under revision at *IEEE Transactions on Affective Computing*.

Nanjing University of Posts and Telecommunications

May 2020 – Oct. 2021

Undergraduate Researcher

Nanjing, China

- **Supervisor:** Prof. Liya Huang
- **Activity:** Investigated EEG-based emergency braking intention detection. Responsibilities included experimental paradigm design, EEG data preprocessing, K-Order Propagation Number feature extraction, and SVM-based classification.
- **Outcomes:** Published one co-first-author paper in *Brain Sciences*.

PUBLICATIONS

Note: Yuan Liao indicates the applicant; * indicates equal-contribution co-first authors.

- **Synergistic material–algorithm design enables robust auditory attention decoding brain-computer interface applications** (*Nature Sensors*, 2026; DOI: 10.1038/s44460-026-00103-6)
Yuzhe Gu*, Yuan Liao*, Jingyu He, Haoxuan Yu, Guansheng Xing, Xiaotian Wang, Aoxi Yu, Qianhe Shu, Liya Huang, Shujun Liu, Keqiang Gao, Wei Huang, Qiang Zhao, et al. **Role: equal-contribution co-first author, second-listed.**
- **Closed-loop correction reprogramming for fine-grained visual prompting** (*Neural Networks*, Volume 202, Article 108991, 2026)
Xueyi Zhang*, Yuan Liao*, Siqi Cai, Mingrui Lao, Haizhou Li. **Role: equal-contribution co-first author, second-listed.** DOI: 10.1016/j.neunet.2026.108991.
- **FD-ARL: Feature Disentanglement with Adversarial-Reconstruction Learning for Cross-Subject Auditory Attention Decoding** (ICASSP 2026 – IEEE International Conference on Acoustics, Speech and Signal Processing)
Yuan Liao, Haoqi Hu, Siqi Cai, Haizhou Li. DOI: 10.1109/ICASSP55912.2026.11461770.
- **Brainprint-Modulated Target Speaker Extraction** (ICASSP 2026 – IEEE International Conference on Acoustics, Speech and Signal Processing)
Qiushi Han*, Yuan Liao*, Youhao Si, Liya Huang. DOI: 10.1109/ICASSP55912.2026.11461698.
- **Frequency-Based Alignment of EEG and Audio Signals Using Contrastive Learning and SincNet for Auditory Attention Detection** (*IEEE Journal of Biomedical and Health Informatics*, under revision)
Yuan Liao, Yuhong Zhang, Rui Dai, Yuanlin Dong, Yu Gu, Yang Yang, Liya Huang.
- **CLDTA: Contrastive Learning based on Diagonal Transformer Autoencoder for Cross-Dataset EEG Emotion Recognition** (*IEEE Transactions on Affective Computing*, under revision; arXiv preprint available)
Yuan Liao, Shuqing Wang, Yuxiang Zhang, Yuhong Zhang, Wei Chen, Yuzhe Gu, Liya Huang.
- **TrustCLIP: Learning from Noisy Labels via Semantic Label Verification and Trust-aligned Gradient Projection** (Proceedings of the 33rd ACM International Conference on Multimedia, 2025)
Xueyi Zhang, Peng Zhu, Yuan Liao, Xiaoqiang Wang, Mingrui Lao, Siqi Cai, Yusheng Guo, Haizhou Li.
- **The effect of speech representations on EEG-based auditory attention detection** (*Pattern Recognition Letters*, 2026)
Haoqi Hu, Yuan Liao, Siqi Cai, Haizhou Li.
- **ASEAF: Attention-SincNet driven EEG-audio fused target speaker extraction network** (*Biomedical Physics & Engineering Express*, 2026)

Y. Yang, Yuan Liao, Q. Han, J. Peng, Liya Huang.

- **GCANet: Enhancing EEG-based Auditory Attention Decoding with Temporal Frequency GCN and Cross Attention Mechanisms** (*Neuroscience*, 2025)
Rui Dai, Yuan Liao, Qiushi Han, Yuanlin Dong, Yang Yang, Liya Huang.
- **Stretchable organic electrochemical transistors for sustained high-fidelity electrophysiology and deep learning-assisted sleep monitoring** (*Matter*, 2025)
Yuncong Pang, Yang Li, Yuzhe Gu, Bing Yu, Zhen Zhu, Xiaotian Wang, Yuan Liao, Liya Huang, Qiang Zhao, et al.
- **A stretchable, ionic conductive, and adhesive patch electrode with ultra-low on-skin impedance for electrophysiological signal recording** (*Science China Information Sciences*, 2025)
Yang Li, Yuzhe Gu, Si Qian, Yuncong Pang, Aoxi Yu, Songwu Zheng, Weixia Xia, Yuan Liao, Shujun Liu, Liya Huang.
- **Microcrack-Enhanced PEDOT:PSS Textile for Rapid-Response, Stable, and Breathable Wearable Sensors** (*ACS Sensors*, 2025)
Yuzhe Gu, Zixuan Zhang, Weilin Liu, Yuan Liao, Wenfei Fei, Yuxiang Zhang, Tao Zhou, Yang Li, Liya Huang, Jiangxin Li.
- **Emotion Recognition of EEG Signals Based on Contrastive Learning Graph Convolutional Model** (*Journal of Neural Engineering*, 2024)
Yiling Zhang, Yuan Liao, Wei Chen, Xiruo Zhang, Liya Huang.
- **EEG-Based Emotion Recognition Using Graph Convolutional Neural Network with Dual Attention Mechanism** (*Frontiers in Computational Neuroscience*, 2024)
Wei Chen, Yuan Liao, Rui Dai, Yuanlin Dong, Liya Huang.
- **Emergency Braking Intention Detect System Based on K-Order Propagation Number Algorithm: A Network Perspective** (*Brain Sciences*, 2021)
Yuhong Zhang*, Yuan Liao*, Yudi Zhang, Liya Huang. **Role: equal-contribution co-first author, second-listed.**

PATENTS

- **Noise-label-based multimodal recognition method, apparatus, device, storage medium, and program product.** Chinese invention patent application CN120726421A, published Sep. 30, 2025. Inventors: Xueyi Zhang, Siqi Cai, Haizhou Li, Haoqi Hu, Yuan Liao.
- **Event-based lip-reading method, apparatus, device, and storage medium.** Chinese invention patent application CN120635994A, published Sep. 12, 2025. Inventors: Xueyi Zhang, Siqi Cai, Haizhou Li, Haoqi Hu, Yuan Liao.
- **EEG-based auditory speech extraction model training method, apparatus, device, and storage medium.** Chinese invention patent application CN120673751A, published Sep. 19, 2025. Inventors: Siqi Cai, Yuan Liao, Xueyi Zhang, Haoqi Hu, Haizhou Li.

AWARD & HONORS

- Huawei Cup: The 20th China Graduate Mathematical Modeling Competition, National Third Prize | 2023
- Second-Class Scholarship for Graduate Students | 2023
- First-Class Scholarship for Graduate Students | 2022
- National Encouragement Scholarship | 2020, 2021
- Third Prize in Jiangsu Province, National College Student Mathematical Modeling Competition | 2020

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, C/C++

Deep Learning Models: Transformers, GNNs, CNNs, Autoencoders, Contrastive Learning, Adversarial Learning

Frameworks & Libraries: PyTorch, MNE-Python, Scikit-learn, NumPy, Pandas

Signal Processing: EEG preprocessing, time-frequency analysis, auditory attention decoding, physiological signal analysis

Languages: Chinese (Native), English (IELTS 6.5)

TEACHING EXPERIENCE

Teaching Assistant, Digital Circuits

NJUPT-Portland State University Joint Program

Fall 2023

Nanjing, China

- Provided instruction and support in an all-English environment for third-year undergraduate students.
- Led 40 lab sessions in assembly language programming, graded assignments, and conducted Q&A sessions.

REFERENCES

1. Prof. Haizhou Li (IEEE Fellow)

School of Artificial Intelligence, School of Data Science, SRIBD, The Chinese University of Hong Kong, Shenzhen, China

Email: haizhouli@cuhk.edu.cn

Relationship: Research Supervisor

2. Prof. Liya Huang

Nanjing University of Posts and Telecommunications

Email: huangly@njupt.edu.cn

Relationship: M.S. Thesis Supervisor

3. Prof. Siqi Cai

School of Intelligence Science and Engineering, Harbin Institute of Technology, Shenzhen, China

Email: caisqi@hit.edu.cn

Relationship: Research Supervisor